

Conversation Task

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Mustafa Özgür Hocaoglu
Computer Engineering Dept.
Marmara University
İstanbul, Turkey
mustafaozgurh@gmail.com

Emir Büçkün
Computer Engineering Dept.
Marmara University
İstanbul, Turkey
emirbuckun@hotmail.com

Osman Buğra Göktaş
Computer Engineering Dept.
Marmara University
İstanbul, Turkey
osmanbugrag@gmail.com

Engin Bilici
Computer Engineering Dept.
Marmara University
İstanbul, Turkey
enginbilici2000@gmail.com

Yiğit Mesut Ak
Computer Engineering Dept.
Marmara University
İstanbul, Turkey
yak@marun.edu.tr

Abstract—

This paper presents the results of our experiments with fine-tuning the Gemma-2b model for conversational response generation. We performed supervised fine-tuning using a dataset of Alpaca-style instructions and evaluated the models using both basic and advanced evaluation metrics. Additionally, we assessed the models' performance on the downstream task of conversational response generation in zero-shot and one-shot settings. Our findings indicate that task-specific fine-tuning significantly improves model performance, and advanced evaluation metrics provide a more comprehensive understanding of model capabilities.

Keywords—*fine-tuning, gemma-2b, conversation, NLP, dataset, evaluation, metrics, downstream*

I. INTRODUCTION

Large Language Models (LLMs) like Gemma-2b have demonstrated significant potential in various natural language processing (NLP) tasks. However, fine-tuning these models for specific tasks and evaluating their performance remains a challenging endeavor. This paper documents our efforts to fine-tune Gemma-2b for conversational response generation and evaluate its performance using both traditional and advanced metrics.

II. METHODOLOGY

A. Dataset

We used a dataset of Alpaca-style instructions for fine-tuning and created a separate test set for conversational response generation.

B. Fine-Tuning

We employed the Unsloth library for fine-tuning, utilizing techniques like 4-bit quantization and Parameter Efficient Fine-Tuning (PEFT) with LoRA.

III. EVALUATION METRICS

We used a combination of basic (ROUGE, BLEU, BERTScore) and advanced (Mover Score, Bleurt, Sentence Bert Score) evaluation metrics.

IV. RESULTS

A. Basic and Advanced Metrics

Our fine-tuned models showed improvements in traditional metrics, with advanced metrics providing deeper insights into model performance.

B. Downstream Task Evaluation

In the zero-shot setting, the model showed moderate performance. However, one-shot fine-tuning with a small labelled dataset significantly improved the results.

V. DISCUSSION

The results underscore the importance of fine-tuning for specific tasks and the value of advanced evaluation metrics in understanding model performance. Task-specific data, even in small amounts, can greatly enhance model capabilities.

VI. CONCLUSION

Our experiments highlight the effectiveness of fine-tuning LLMs like Gemma-2b for conversational response generation. Advanced evaluation metrics and downstream task evaluations provide a comprehensive picture of model performance. Future work will explore more extensive fine-tuning and evaluation strategies to further optimize model performance.

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